Let $I=(a,f)\subseteq R[x]$ be an ideal generated by $a, f\in R[x]$.

Consider the diagram:

$$R[x] \xrightarrow{\varphi_a} (R/aR)[x] \xrightarrow{\pi_{\bar{f}}} (R/aR)[x]/(\bar{f}).$$

$$R[x] \xrightarrow{\pi_a} R[x]/aR[x] \xrightarrow{\simeq} (R/aR)[x]$$

$R[x]\to R[x]/aR[x]$
: $P(x)\mapsto P(x)+aR[x]$
Onto Homomorphism.

$R[x]/aR[x]\to (R/aR)[x]$
: $P(x)+aR[x]\mapsto \overline{P(x)}$, where $\overline{\sum a_n\cdot x^n}=\sum(a_n+aR)\cdot x^n$.
Isomorphism.

$\Rightarrow \varphi: R[x]\to (R/aR)[x]$ is Onto Homomorphism, So is $\pi_{\bar{f}}\circ\varphi$ onto Homomorphism.

Moreover,

$$\ker(\pi_{\bar{f}}\circ\varphi)=(\pi_{\bar{f}}\circ\varphi)^{-1}[\{0\}]=\varphi^{-1}[\ker\pi_{\bar{f}}]=\varphi^{-1}[(\bar{f})]$$
$$=\{P(x)\in R[x]\mid \overline{P(x)}\in\ker\pi_{\bar{f}}\}=\{P(x)\in R[x]\mid \overline{P(x)}\in(\bar{f})\}$$
$$\overset{(*)}{=}\{P(x)\in aR[x]+f(x)R[x]\}=(a)+(f)=(a,f)=I.$$

Assume $a\in R$ and $(a)\subseteq R$ is maximal, and $\bar{f}\in(R/aR)[x]$ is irreducible.
Then, R/aR is a field, and therefore $(R/aR)[x]$ is P.I.D.
So, $(\bar{f})$ is maximal, it implies that $(R/aR)[x]/(\bar{f})$ is a field.
Now, by the first-isomorphism theorem,

$$R[x]/(aR[x]+fR[x])\cong(R/aR)[x]/\bar{f}\cdot(R/aR)[x]$$

Note: $(*)$ is obtained by: given $P(x)\in R[x]$,

$$\overline{P}(x)\in(\overline{f(x)})\iff \overline{P(x)}=\overline{g(x)}\cdot\overline{f(x)} \text{ for some } g(x)\in R[x]$$
$$\iff P(x)-g(x)f(x)\in aR[x] \text{ for some } g(x)\in R[x]$$
$$\iff P(x)\in(a)+(f(x))=(a,f).$$

Investigate $\mathbb{Z}[x]/(6, x^2+x+1)$

1). $(6, x^2+x+1)$

(-1). x^2+x+1 is irreducible.

x^2+x+1 reducible $\Rightarrow x^2+x+1=(x+a)\cdot(x+b)$ s.t. $ab=1$

So $ab=1 \Rightarrow a=b=1$ or $a=b=-1$, but ± 1 are not roots,

(-2). $(6, x^2+x+1) = (2, x^2+x+1) \cap (3, x^2+x+1)$

Denote $f = x^2+x+1$. Then,

$$(6,f) = (6)+(f) = (2)(3)+(f) = [(2)+(f)]\cdot[(3)+(f)] = (2,f)\cdot(3,f) = (2,f)\cap(3,f)$$

Thus, by CRT,

$$\mathbb{Z}[x]/(6,f) \cong \left(\mathbb{Z}[x]/(2,f)\right) \times \left(\mathbb{Z}[x]/(3,f)\right).$$

2). Consider

$$\mathbb{Z}[x] \to (\mathbb{Z}/(2))[x] \to (\mathbb{Z}/(2))[x]/(\overline{f})$$

$$\sum a_n x^n \mapsto \sum (a_n \bmod 2) x^n \mapsto \sum (a_n \bmod 2) x^n + (\overline{f})$$

which is Onto Homomorphism with kernel $(6,f)$.

Since $\overline{f} = \overline{x}^2+\overline{x}+1$ has no root in $\mathbb{F}_2 \cong \mathbb{Z}/(2)$, so irreducible, thus

$$\mathbb{Z}[x]/(2,f) \cong \mathbb{F}_2[x]/(\overline{x}^2+\overline{x}+1) \cong \mathbb{F}_{2^2}.$$

But, $\overline{f}$ is not irreducible over $\mathbb{Z}/(3) \cong \mathbb{F}_3$, as $\overline{x}^2+\overline{x}+1 = (\overline{x}+2)^2$, so

$$\mathbb{Z}[x]/(3,f) \cong \mathbb{F}_3[x]/(\overline{x}^2+\overline{x}+1)$$

By the division algorithm, given $f(x) \in \mathbb{Z}[x]$,

$$f(x) = b(x)\cdot(x^2+x+1) + ax + b \quad \text{for some } b(x) \in \mathbb{Z}[x] \text{ and } a, b \in \mathbb{Z},$$

So that $\mathbb{F}_3[x]/(\overline{x}^2+\overline{x}+1) = \{ax+b+(\overline{x}^2+\overline{x}+1) \mid a, b \in \mathbb{F}_3\}$.

$(a\overline{x}+b)\cdot(c\overline{x}+d) = ac\,\overline{x}^2 + (ad+bc)\cdot\overline{x} + bd = (2ac+ad+bc)\overline{x} + ac + bd$

If $a=0$, then b is an element of a field $\mathbb{F}_3$, so it is a unit unless it is zero.

If $a=1$, then $(2c+d+bc)\cdot\overline{x} + c + bd = 1$